

Precise Definitions and Counterexamples

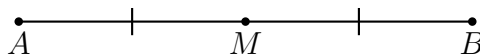
- A definition in geometry has to be *testable*: every condition it names must be one you could check on a figure. If a condition is missing, some figure will satisfy the words and still be the wrong thing — that figure is a **counterexample**.
- Each diagram shows only the labelling. All measurements are given in the problem text.
- Show the equation you wrote before you solve it.

1. Three students try to define the *midpoint* of $\overline{AB}$.

Attempt A	“A point that is the same distance from A as it is from B .”
Attempt B	“The point of $\overline{AB}$ that divides it into two congruent segments.”
Attempt C	“The point where the two halves of the segment look equal.”

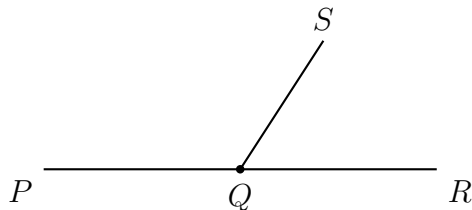
Which attempt is a precise definition? For each of the other two, name the condition it leaves out or the word that cannot be tested.

2. M is the midpoint of $\overline{AB}$, as marked below. In centimetres, $AM = 3x - 4$ and $MB = x + 6$. Use the definition of midpoint to write an equation, then find x .



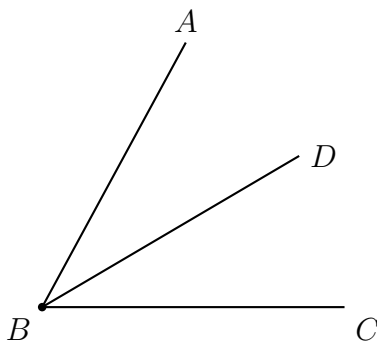
Answer: _____

3. A student claims: “If two angles have measures that add to 180° , then they form a linear pair.” The figure shows what a linear pair actually looks like: two adjacent angles whose non-common sides are opposite rays.



Give a counterexample to the student’s claim. State the two angle measures you choose and explain why your pair is not a linear pair.

4. $\overrightarrow{BD}$ bisects $\angle ABC$, as marked. In degrees, $m\angle ABD = 5x + 2$ and $m\angle DBC = 7x - 14$.



- (a) Use the definition of an angle bisector to write an equation and find x .

Answer: _____

- (b) Find $m\angle ABC$.

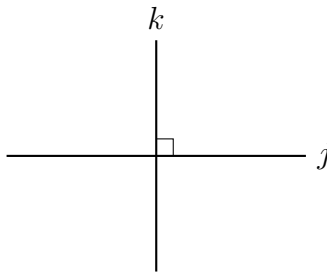
Answer: _____ degrees

5. A student writes: “*Perpendicular lines are two lines that intersect.*”

(a) That definition is not precise. Explain what a figure could look like that fits the words and is still not a pair of perpendicular lines, then state the condition that must be added.

Answer: _____

(b) Lines j and k are perpendicular. One of the four angles they form measures $(4x + 10)^\circ$. Use the corrected definition to find x .



Answer: _____

6. Give a counterexample to each claim. A counterexample must satisfy the “if” part and fail the “then” part, so say clearly why yours does both.

(a) “If two angles are congruent, then they are vertical angles.”

(b) “If a point is the same distance from A as it is from B , then it is the midpoint of $\overline{AB}$.”

7. $\overline{PQ} \cong \overline{RS}$. In centimetres, $PQ = 2x + 1$ and $RS = 5x - 14$.



(a) Use the definition of congruent segments to write an equation and find x .

Answer: _____

(b) Find the length PQ .

Answer: _____ cm

8. A student proposes this definition: *“The midpoint of a segment is a point on the segment that divides it into two segments.”*

(a) Give a counterexample: describe a point that fits every word of this definition and is not the midpoint. Use lengths in your description.

(b) Rewrite the definition so that your counterexample no longer fits it. List every condition your version requires.